

Chapter 1

Taylor's Theorem

This chapter is motivated by the question: how can we approximate a function with other, simpler functions, locally? In particular, the approximation of functions by *polynomials*. Polynomials are the natural first choice as they are, in some way, the simplest type of function on $\mathbb{R}$ and have very nice properties.

When discussing Taylor's theorem, we are fundamentally talking about *polynomials*, not power series. A series that defines a function is not just a single series; for every x , you have a different limit. Establishing whether such a series converges, is continuous, or is differentiable requires significant machinery in general (we will see this in the sequences and series of functions topic), but for functions with a Taylor Series, these properties are automatic because of the properties of polynomials.

1.1 Taylor Polynomials

Definition 1.1 (Taylor Polynomial). Let f be n times differentiable at a point a . The n -th order *Taylor polynomial* of f centred at a is:

$$P_{n,a}(x) := \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k.$$

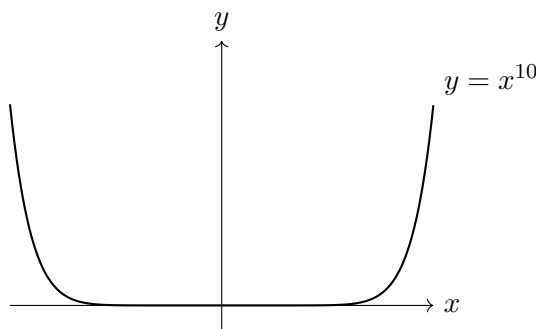
Remark. We can construct a polynomial with prescribed derivatives at a point a by writing it in “centred” form $\sum_{k=0}^n c_k (x-a)^k$. Then the derivatives of this polynomial at a directly determine the coefficients $c_k = f^{(k)}(a)/k!$.

Geometrically, by forcing the polynomial and the function to agree up to n derivatives, the *local geometry* of the two functions becomes identical up to order n . To make this more clear, consider the remainder function below:

Definition 1.2 (The Remainder). The *remainder* of the Taylor approximation of order n is the difference between the function and its Taylor polynomial:

$$R_{n,a}(x) := f(x) - P_{n,a}(x).$$

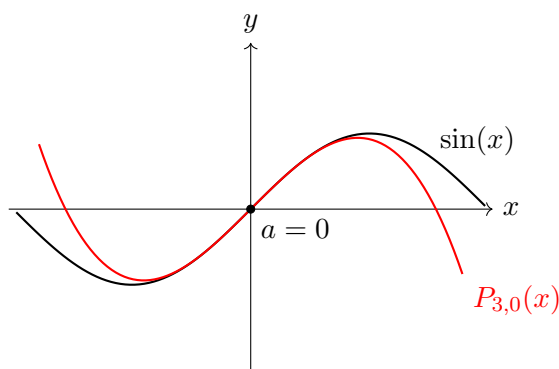
Remark. The remainder represents the “error” in the approximation of f by $P_{n,a}$. Notice that the remainder $R_{n,a}$ has its first n derivatives at a vanishing, which geometrically corresponds to extreme “flatness” at a , showing that the local approximation near a is extremely good. This is illustrated below with x^{10} at 0; there is a 10 multiplicity root at 0, which geometrically corresponds to extreme “flatness” at zero.



In terms of the polynomial approximation to a function, the above “flatness” corresponds to the polynomial “hugging” the function tightly, locally near the point a (although, of course, the remainder is not necessarily a polynomial like x^{10} , but the flatness about a mimics that of a polynomial with the appropriate root multiplicity there). More conceptually, this phenomenon is explained by the limit

$$\lim_{x \rightarrow a} \frac{R_{n,a}(x)}{(x-a)^k} = 0$$

for $k \leq n$; i.e. the R “dominates” the behaviour compared to any lower degree polynomial with root multiplicity $k < n$ at that point. We illustrate this below with the example of $f(x) = \sin x$ and the corresponding 3-rd order Taylor polynomial $P_{3,0}(x) = x - \frac{x^3}{6}$. Notice that the approximation no longer works for points away from a ; this is expected, as $\sin x$ is not a polynomial (though it does have a power series, as we will see later).



Definition 1.3 (Little- o Notation). Let $k \in \mathbb{N}$. We say that $f \in o(|x - a|^k)$ if

$$\lim_{x \rightarrow a} \frac{f(x)}{|x - a|^k} = 0.$$

Example 1.4. As we explained above, $R_{n,a} \in o(|x - a|^n)$.

Theorem 1.5 (Taylor's Theorem with Lagrange Remainder). *Let f be $n + 1$ times differentiable on an open interval containing a and x . Then there exists some θ strictly between a and x such that:*

$$R_{n,a}(x) = \frac{f^{(n+1)}(\theta)}{(n+1)!} (x - a)^{n+1}.$$

Proof. Fix $x \neq a$. Consider the function $F(t)$ defined on the closed interval between a and x :

$$F(t) := f(x) - \sum_{k=0}^n \frac{f^{(k)}(t)}{k!} (x - t)^k.$$

Notice that $F(x) = 0$ and $F(a) = R_{n,a}(x)$. Differentiating $F(t)$ with respect to t causes the sum to telescope, leaving only the final term:

$$F'(t) = -\frac{f^{(n+1)}(t)}{n!} (x - t)^n.$$

Now, define an auxiliary function $G(t)$ to apply Rolle's Theorem:

$$G(t) := F(t) - \left(\frac{x-t}{x-a} \right)^{n+1} F(a).$$

We have $G(x) = 0$ and $G(a) = F(a) - F(a) = 0$. Since G is continuous on the closed interval between a and x and differentiable on the open interval, Rolle's Theorem guarantees a point θ strictly between a and x where $G'(\theta) = 0$.

Differentiating $G(t)$ gives:

$$G'(t) = F'(t) + (n+1) \frac{(x-t)^n}{(x-a)^{n+1}} F(a).$$

Setting $G'(\theta) = 0$ and substituting $F'(\theta)$ yields:

$$\frac{f^{(n+1)}(\theta)}{n!} (x-\theta)^n = (n+1) \frac{(x-\theta)^n}{(x-a)^{n+1}} R_{n,a}(x).$$

Cancelling $(x-\theta)^n$ (since $\theta \neq x$) and rearranging gives the desired form for $R_{n,a}(x)$. $\square$

Remark. The Lagrange form is incredibly convenient for rough approximations. If the $(n+1)$ -th derivative is bounded, the remainder is immediately bounded. However, it has significant drawbacks: the value θ depends on x in a way that is not explicitly understood and is generally not continuous. Therefore, if we want to differentiate the remainder or evaluate it precisely, the Lagrange form is inadequate.

Theorem 1.6 (Taylor's Theorem with Integral Remainder). *Let f be $n+1$ times continuously differentiable on an open interval containing a . For any x in this interval,*

$$R_{n,a}(x) = \int_a^x \frac{f^{(n+1)}(t)}{n!} (x-t)^n dt.$$

Proof. We proceed by mathematical induction on n . For $n = 0$, the theorem states $f(x) - f(a) = \int_a^x f'(t) dt$, which is exactly the Fundamental Theorem of Calculus.

Assume the formula holds for some k . We apply integration by parts to the integral remainder for k . Let $u = f^{(k+1)}(t)$ and $v = -\frac{(x-t)^{k+1}}{k+1}$. Then $du = f^{(k+2)}(t) dt$ and $dv = (x-t)^k dt$.

$$\int_a^x \frac{f^{(k+1)}(t)}{k!} (x-t)^k dt = \left[-f^{(k+1)}(t) \frac{(x-t)^{k+1}}{(k+1)!} \right]_a^x + \int_a^x \frac{f^{(k+2)}(t)}{(k+1)!} (x-t)^{k+1} dt.$$

Evaluating the boundary term at x yields 0. Evaluating at a yields $\frac{f^{(k+1)}(a)}{(k+1)!}(x-a)^{k+1}$. Thus,

$$R_{k,a}(x) = \frac{f^{(k+1)}(a)}{(k+1)!}(x-a)^{k+1} + \int_a^x \frac{f^{(k+2)}(t)}{(k+1)!}(x-t)^{k+1} dt.$$

Since $R_{k,a}(x) - \frac{f^{(k+1)}(a)}{(k+1)!}(x-a)^{k+1} = R_{k+1,a}(x)$, the result holds for $n = k+1$. $\square$

Remark. The integral remainder provides an *explicit* formula for the remainder that works for every x . Because the dependence on x is explicitly captured inside the integral (and not hidden in an arbitrary θ), we can use the Fundamental Theorem of Calculus to differentiate the remainder, bound it precisely, and evaluate it anywhere.

1.2 Taylor Series

We have seen that Taylor polynomials are fundamentally local approximations, however there is a special class of functions, called *analytic functions*, whose Taylor polynomials converge to the original function on an interval as $n \rightarrow \infty$. This is quite surprising, as it says that these functions can be recovered *globally* in the form of an “infinite polynomial” from only their derivatives *at a single point*, i.e. from only local information. Indeed, these *are* the special functions defined by this property; there are many examples, as we will see below, of functions that are in C^∞ (infinitely differentiable) but are not analytic.

Definition 1.7 (Real Analytic Function). We say that a function $f \in C^\infty$ is *real analytic* at a point a if there exists $R > 0$ such that for every $x \in (a-R, a+R)$ we have $\lim_{n \rightarrow \infty} P_{n,a}(x) = f(x)$. In this case, we write

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

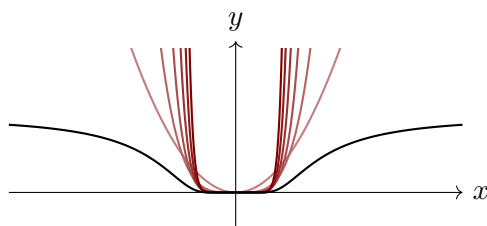
The maximal $R > 0$ for which this property holds is called the *radius of convergence* of the Taylor series.

Intuitively, this means that the function is “rigid” enough that it can be approached globally (i.e. on an interval) by polynomial functions. This is precisely why the derivatives of the function at one point generate can build the whole function; as we saw in the discussion at the start of this chapter, polynomials are extremely special functions that can be re-centred to any point while remaining invariant, and so the local information of an analytic

function inherits this property, such that local information, like derivatives at a point a , can still recover the function globally.

Example 1.8. Not all functions in C^∞ are (real) analytic. A classic counterexample is the bump function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} \exp(-1/x^2), & x \neq 0, \\ 0, & x = 0. \end{cases}$$



This function is infinitely differentiable everywhere (which one should verify), but $f^{(n)}(0) = 0$ for all $n \in \mathbb{N}$. Therefore, its Taylor polynomial at $a = 0$ is the identically zero function, $P_{n,0}(x) = 0$ for all n .

Intuitively, because this function converges to zero faster than any polynomial, on any interval around 0, each function x^n is “too weak” to match the exponential decay of f near 0, so has nothing to contribute to the Taylor polynomial at 0. In other words, it is because $f \in o(|x|^n)$ for every $n \in \mathbb{N}$. There is a complete loss of information, demonstrating that Taylor polynomials are fundamentally a *local* matching tool, apart from in the special case of analytic functions.

Based on the Lagrange form of the remainder in Taylor's Theorem, we get a useful sufficient condition for the convergence of Taylor series to the original function.

Theorem 1.9. Let $f \in C^\infty$ be defined on $(a - \rho, a + \rho)$ for some $\rho > 0$. If there exists $C, R > 0$ and $N \in \mathbb{N}$ such that for all $x \in (a - \rho, a + \rho)$ and all $n \geq N$,

$$|f^{(n+1)}(x)| \leq C \frac{(n+1)!}{R^{n+1}},$$

then the Taylor polynomial $P_{n,a}(x)$ of f converges to $f(x)$ as $n \rightarrow \infty$ for every x satisfying

$$|x - a| < \min\{\rho, R\}.$$

In particular, f is analytic on $|x - a| < \min\{\rho, R\}$.

Proof. Using the Lagrange form of the remainder (Theorem 1.5), we have, for x such that $|x - a| < \min\{\rho, R\}$,

$$R_{n,a}(x) = \frac{f^{(n+1)}(\theta)}{(n+1)!} (x - a)^{n+1},$$

for some θ between a and x . Since $|x - a| < \min\{\rho, R\}$, and θ is between a and x , we also have $|\theta - a| < \min\{\rho, R\}$, so for $n \geq N$, the derivative bound applies at θ also:

$$|R_{n,a}(x)| = \left| \frac{f^{(n+1)}(\theta)}{(n+1)!} (x-a)^{n+1} \right| \leq \frac{C \frac{(n+1)!}{R^{n+1}}}{(n+1)!} |x-a|^{n+1} = C \left(\frac{|x-a|}{R} \right)^{n+1}.$$

Since, in particular, $|x - a| < R$ we know $\frac{|x-a|}{R} < 1$. Therefore, as $n \rightarrow \infty$, the remainder $|R_{n,a}(x)| \rightarrow 0$. This proves that the sequence of Taylor polynomials converges pointwise to $f(x)$. $\square$

Remark. The expression $\frac{(n+1)!}{R^{n+1}}$ is highly advantageous when bounding error because factorials grow significantly faster than exponentials. This affords us a massive amount of “room” to bound the derivatives of functions like $\sin(x)$, $\cos(x)$, and $\exp(x)$, which are indeed analytic functions.

Exercise. Prove that $\exp(x)$ is real analytic.